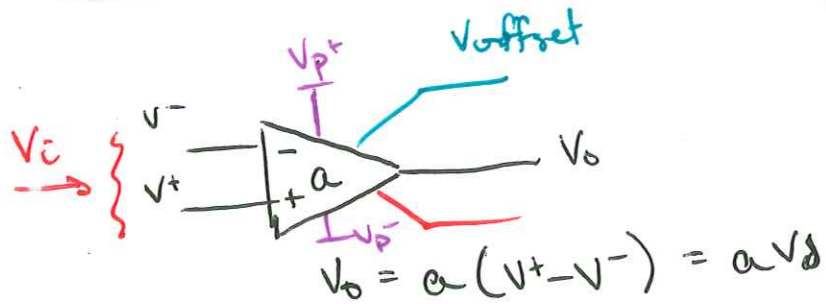


Op Amps

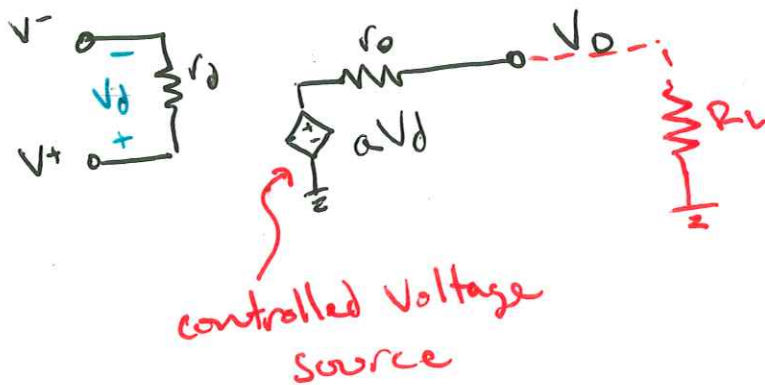


I want:



$$V_o = AV_i \rightarrow \frac{V_o}{V_i} = A$$

"Ideal" Op Amp Model



- r_o - output impedance
- r_d - Differential input impedance
- V_d - Diff. input voltage.
- a - ~~closed~~ open-loop, "internal" gain.

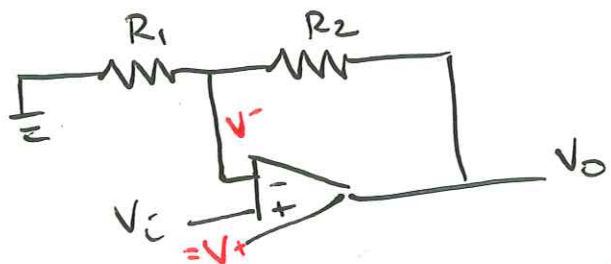
ideally, $a \rightarrow \infty$
 realistically, $a \approx 10^5 - 10^7 \text{ V/V}$
 OP27: $a \approx 10^6 \text{ V/V} = 1 \text{ V/mV}$
 741: $a \approx 10^5 \text{ V/V}$

"Basic" Analysis

$$r_d \rightarrow \infty, r_o \rightarrow 0, a \neq \infty$$

A = Closed-Loop Gain

Non-Inverting Amp. using Basic Analysis



feedback gain

voltage Divider!

$$V^- = \frac{R_1}{R_1 + R_2} V_o = b V_o$$

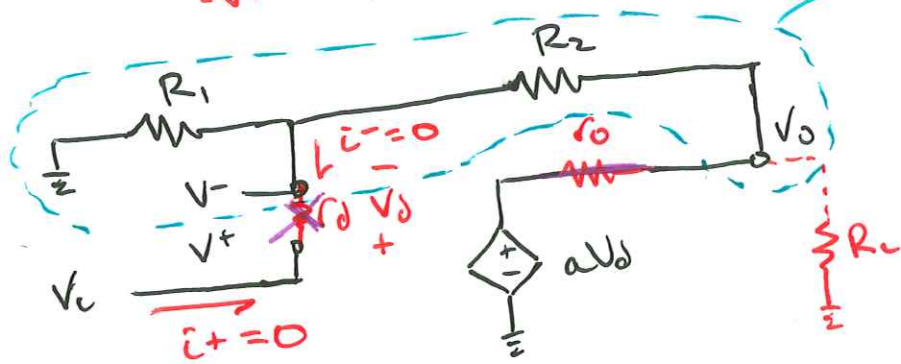
$$V^+ = V_i$$

$$V_d = V^+ - V^- = V_i - b V_o$$

$$V_o = a V_d$$

$$V_o = a(V_i - b V_o)$$

$$\frac{V_o}{V_i} = \frac{a}{1 + ab} = A \rightarrow \text{closed loop gain.}$$



$$r_d = \infty$$

$$r_o = 0$$

$$a \neq \infty$$

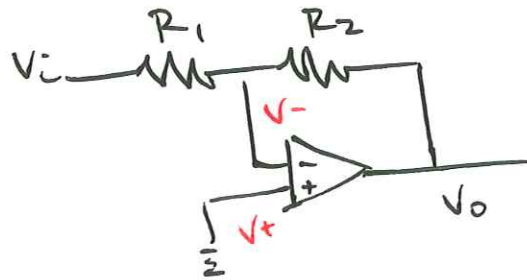
$$\frac{V_o}{V_i} = \frac{1}{\frac{1}{a} + b} = \frac{1}{b} \frac{1}{1 + 1/ab}$$

what if $a \rightarrow \infty$

$$\frac{V_o}{V_i} = \frac{1}{b} = A_{ideal}$$

$$\frac{R_1 + R_2}{R_1} = 1 + \frac{R_2}{R_1} = \frac{1}{b}$$

Inverting Amp



$$V^+ = 0$$

$$V_o = a(V^+ - V^-) = -aV^- \quad (1)$$

$$V_n = (V_o - V_i) \frac{R_1}{R_1 + R_2} + V_i$$

$$V_n = \frac{R_1}{R_1 + R_2} V_o + V_i \left(1 - \frac{R_1}{R_1 + R_2}\right) \quad (2)$$

$$\textcircled{1} \textcircled{2} \quad \frac{V_o}{V_i} = A = - \frac{R_2}{R_1} \left(\frac{1}{1 + (1 + R_2/R_1)/a} \right)$$

if $a \rightarrow \infty$

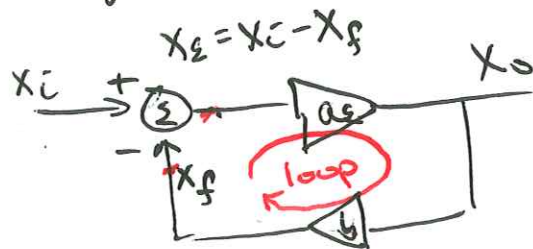
$$\frac{V_o}{V_i} = - \frac{R_2}{R_1} = A_{ideal} = \frac{1}{b}$$

b in this case not directly the Voltage Div. Gain!

want $\frac{V_o}{V_i} = A_{ideal} \frac{1}{1 + 1/a b}$ type form!

cannot do with this case however...

Negative feedback Systems



$a_e \rightarrow$ Error Amplifier (ideal $a_e \rightarrow \infty$)

$$L = a_e b = \text{loop Gain}$$

for negative feedback,

$$L \geq 0$$

$$x_f = b x_o$$

$$x_e = x_i - b x_o \sim v_d = v^+ - v^-$$

$$x_o = a_e x_e \sim v_o = a(v^+ - v^-)$$

$$\frac{x_o}{x_i} = \frac{a_e}{1 + a_e b} = A_{ideal} \frac{1}{1 + 1/a_e b}$$

$$A_{ideal} = \frac{1}{b}$$

$$= A_{ideal} \frac{1}{1 + 1/L} = A_{ideal} \left[1 - \frac{1}{1+L} \right]$$

$$0 \leq \frac{1}{1+L} \leq 1$$

Fractional Departure from ideal

Non-Inverting:

$$b = \frac{R_1}{R_1 + R_2}, \quad L = a \frac{R_1}{R_1 + R_2}$$

$$a_e = a$$

Inverting:

$$b = -\frac{R_1}{R_2}, \quad L = a / (1 + R_2/R_1) = a_e b = -\frac{R_1}{R_2} a_e$$

$$a_e = -\frac{a R_2}{R_1 + R_2}$$